Dual ordinal spectral ranking of journals and institutions

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Abstract

We develop a dual spectral ranking of journals and institutions that jointly determines prestige scores via a Katz resolvent applied to a block attention matrix constructed from intra-field citation flows. A continuous parameter χ apportions each reference between source and institution channels; a binary block-mask selects among four bibliometrically distinct flow structures, nesting conventional single-mode rankings as special cases. Applied to the economics and commerce literature using OpenAlex data, the method produces stable, interpretable joint rankings whose sensitivity to model parameters is examined systematically.

Keywords: spectral ranking, dual journal–institution prestige, Katz-Hubbell resolvent

1 Introduction

The research system relies on journals and institutions. Journals filter works through editorial and peer review, archive the literature, and create metadata for retrieval and exploration. Institutions fund individual researchers, implement training, oversee compliance, establish research organisational units and maintain physical and information infrastructure. Although the status hierarchy that links journals and institutions is a pervasive aspect of the research system (Merton, 1973; McNamee and Willis, 1994; Martin, 1998), bibliometric studies that explore the joint hierarchy of journals and institutions are rare. This paper proposes a joint ordinal spectral ranking of journals and institutions derived from bibliometric data. We hope that the method helps to reveal how status is formed and maintained. We also intend to apply the approach to study the relations between the status of journals and institutions, and researcher esteem.

The introduction of the science citation index (Garfield, 1964) opened a new way to investigate connections between journals and institutions. Citation-based journal ranking in particular became a useful means to manage collections and to select outlets for publications. On the other hand, the misapplication of the journal's impact factor (JIF) as proxy for the importance of research published in that journal fostered a still-unresolved tension in bibliometric evaluation (Garfield, 1979; MacRoberts and MacRoberts, 2018). Perhaps due to the complexity of disciplines and missions, an institution impact factor (IIF) has not emerged. Rather, institutional rankings derived from bibliometric indicators of works, authors or journals are often composed with other indicators to form an institutional score. This is the approach in global rankings such as QS, Times Higher Education (THE) and Shanghai Ranking (ARWU), the academic Leiden and SCImago Institutions rankings, as well as many national performance-based research funding schemes (Zacharewicz et al, 2019).

The prospects for using a citation index to explore research importance and usage measures were considered in an early work by Narin (1976), Pinski and Narin (1976), and Narin et al (1976) who proposed a recursive method to construct influence weights of journals within a disciplinary field. To obtain influence weights for institutionszz summed the publications in the disciplinary journal sets, weighted by the respective journal influence score. Over a sample of several dozen universities, the total institution influence scores were well-correlated with survey-based rankings, butz also with the number of publications, suggesting that survey respondents combined influence and size. Narin's work built on earlier studies of social networks and developed into the eigenvector approach now widely used for journal ranking (Bergstrom et al, 2008; Vigna, 2016; Franceschet, 2010). Eigenvector methods have also been applied to author-level influence weightings in the field of social science (West et al, 2013), and extended by addition to ranking institutions and countries. An alternative approach iterates between the article-article network linked by the reference lists (without projection) and the corresponding article-author network (Ujum et al, 2015; Pal and Ruj, 2015). The formulation by Cao et al (2023) adopts a similar network, with reference-based article-article links, and the related article-institution links, ranked by a novel and effective iteration over random walks. This is one of very few spectral rankings of institutions. The authors note that a limiting case of this approach corresponds to the eigenvector (pageRank) method. As noted earlier by Geller (1978), random walks over Markov chains are closely related to the spectral ranking problem discussed in this paper.

It is useful to set out some analytical devices that we adopt in this paper. First, we clarify the terms reference and citation used here. A reference is an entry w_2 in the reference list of a work w_1: w_1 references w_2 and w_2 is referenced by w_1. Bibliometric indexers establish the connection between w_2 and the entry for w_2 in a list of works, and may publish an index of all works that reference w_2 (Wouters, 1999). With this indexing w_1 cites w_2 and w_2 is cited by w_1. A reference can be fractionated among its authors and institutions. We say that a reference signifies the attention given (paid) by w_1 to w_2; in spectral ranking attention may be weakened or strengthened by the attention that has been given by others to w_1. We can also say that a citation signifies the influence of w_2 on w_1; like attention, influence can be weakened or strengthened by other citations to w_2. Second, in this paper we view references and citations through Social Systems Citation Theory (Tahamtan and Bornmann, 2022, SSCT). We concentrate on the structural features of the network of references, and set aside authorial intent. This approach contrasts with actor-network theory (Latour, 2007, ANT), which foregrounds the role of heterogeneous agents in assembling citation networks. ANT will be important in our follow-up studies that relate this paper to the referencing activities of individual researchers.

This paper proceeds as follows. In the next section, we construct a framework for computing the attention or influence of journals and institutions coupled by the reference lists of works they publish. We parametrise many assumptions and variations, so we can assess their consequences in our examples. We then construct a corpus of works in the fields of Economics and Business, and present spectral rankings of journals and institutions in these fields alongside sensitivity studies of many parameters. We then discuss the results, including a critique of our methods, and close by mentioning some directions for future work.

2 Journal–institution citation networks

We begin by selecting N_s sources $\mathfrak{S}$ in a field, discipline or category. Then we select N_w works $\mathfrak{W}$ published in $\mathfrak{S}$ within the census window years $t_c = [t^c_0 \dots t^c_1]$ or target years $t_t = [t^t_0 \dots t^t_1]$. From the reference lists of the set of census works $\mathfrak{W}^c$ we identify the set of target works $\mathfrak{W}^t$ and form the filtered reference set $\mathfrak{R} = \{(i,j) : w_i \in \mathfrak{W}^c,\, w_j \in \mathfrak{W}^t\}$. Author affiliations in $\mathfrak{W}$ establish a set of authors $\mathfrak{A}'$ and institutions $\mathfrak{U}'$. In practice many institutions (and sources) publishing only a few works per year can be dropped from the corpus. Let $\bar{w}_u$ be the mean annual census-window work count for institution u and $\bar{w}_s$ the corresponding count for source s. We determine lower limits τ_U and τ_S by examining the respective frequency distributions and define $\mathfrak{U} \subseteq \mathfrak{U}'$ as the institutions with $\bar{w}_u \geq \tau_U$, and $\mathfrak{S} \subseteq \mathfrak{S}'$ as the sources with $\bar{w}_s \geq \tau_S$, with corresponding author set $\mathfrak{A} \subseteq \mathfrak{A}'$. Works that lose all their authorships under the institution filter, or whose source is dropped by the source filter, are removed from the corpus. Each retained work w_i has an *institution weight* under author-fractional counting (Waltman and van Eck, 2015),

$$\omega_{iu} \;=\; \sum_{\ell:\, u \in \sqcap_{i\ell}} \frac{1}{a_i}\,\frac{1}{u_{i\ell}}, \tag{1}$$

where ℓ indexes the retained authors of work w_i, a_i is their count, and $u_{i\ell}$ the number of retained institutions of author ℓ, so that $\sum_u \omega_{iu} = 1$. Re-weighting fractional counts as a consequence of pruning the number of institutions is a trade-off between boosting a work's remaining institutions and diminishing the work's total contribution.

When a work w_i references another work w_j we say that w_i gives attention to w_j. The amount of attention given by a work' references can be either (a) the number of references, or (b) a fixed amount (Zitt and Small, 2008). Let $R_i = |\{j : (i,j) \in \mathfrak{R}\}|$ be the reference count of work w_i and $\bar{R} = |\mathfrak{R}|/N_w$ the mean reference count over $\mathfrak{W}$. We use the *reference normalisation weight* ρ_i to select case (a) or case (b) respectively:

$$\rho_i \;=\; \begin{cases} 1 & \text{(full count)} \\ \bar{R}/R_i & \text{(fixed count).} \end{cases} \tag{2}$$

We construct the set of *units* [sources or institutions; Pinski and Narin (1976)] $\mathfrak{P} = \mathfrak{S} \cup \mathfrak{U}$ with $N = N_s + N_u$ elements, ordered so that sources occupy indices $1, \ldots, N_s$ and institutions indices $N_s+1, \ldots, N$. A reference $(i, j) \in \mathfrak{R}$ gives attention across four unit-type pairs: source–source (SS), source–institution (SI), institution–source (IS), and institution–institution (II).

For each work w_i we define the source membership one-hot vector $\mathbf{b}_i^S \in \mathbb{R}^{N_s}$ with $[\mathbf{b}_i^S]_s = \mathbf{1}[w_i \in s]$, and the institution weight distribution vector $\mathbf{b}_i^U \in \mathbb{R}^{N_u}$ with $[\mathbf{b}_i^U]_u = \omega_{iu}$ from (1). Attention transfer between pairs of unit types is accumulated as four blocks over $\mathfrak{R}$ as rank-1 outer products,

$$\mathbf{C}_{XY} \;=\; \sum_{(i,j) \in \mathfrak{R}} \rho_i \, \mathbf{b}_i^X \left(\mathbf{b}_j^Y\right)^\top, \qquad XY \in \{SS,\, SI,\, IS,\, II\}, \tag{3}$$

with $\mathbf{b}_i^X$ weighting the referencing (citing) side and $\mathbf{b}_j^Y$ the referenced (cited) side.

The citation matrix assembles the four blocks:

$$\mathbf{C}(\chi, \mathbf{m}, \rho) \;=\; \begin{bmatrix} m_{SS}(1-\chi)^2\, \mathbf{C}_{SS} & m_{SI}\, \chi(1-\chi)\, \mathbf{C}_{SI} \\ m_{IS}\, \chi(1-\chi)\, \mathbf{C}_{IS} & m_{II}\, \chi^2\, \mathbf{C}_{II} \end{bmatrix}, \tag{4}$$

where the binary *block mask* $\mathbf{m} = (m_{SS}, m_{SI}, m_{IS}, m_{II}) \in \{0,1\}^4$ is bibliometrically meaningful for source-only (1000), institution-only (0001), bipartite (0110), and full-joint (1111) combined blocks. The mixing weight χ controls the source–institution balance in the bipartite and full-joint cases and is inert in the single-mode cases. The χ scaling preserves total citation flow across all block combinations, since $(1-\chi)^2 + 2\chi(1-\chi) + \chi^2 = 1$. We could name $\mathbf{C}$ the reference, attention or citation matrix and prefer the latter.

Note that the corpus excludes (a) census work references not giving attention to target works, (b) non-census works giving attention to target works, and (c) authorships that include an omitted institution.

Ordinal spectral ranking. Let $r_p = \sum_q C_{pq}$ be the total weighted citations from unit p, and $\mathbf{D}_r = \operatorname{diag}(r_p)$ the corresponding diagonal matrix. Let $\boldsymbol{\mu}$ be a prior distribution (boundary condition; Hubbell, 1965; Vigna, 2016), with $\mu_p = 1/N_{\mathfrak{P}}$ for $p \in \mathfrak{P}$. The row-stochastic citation matrix is $\mathbf{H} = \mathbf{D}_r^{-1}\mathbf{C}$. If $r_p = 0$ (a dangling unit with no outgoing citations) the corresponding row of $\mathbf{C}$ is zero and $\mathbf{H}$ is defined to have a zero row too.

The spectral ranking $\boldsymbol{\pi}$ (see Vigna, 2016, for an overview) is defined as the solution to the damped eigenvector problem

$$\boldsymbol{\pi} \;=\; \alpha \mathbf{H}^\top \boldsymbol{\pi} + (1-\alpha)\boldsymbol{\mu}, \qquad \mathbf{1}^\top \boldsymbol{\pi} = 1, \quad \boldsymbol{\pi} \geq 0, \tag{5}$$

with the factor $\alpha \in (0,1)$. As explained by Vigna (2016), α can be regarded as a damping factor applied to walks over the citation network (Katz, 1953) or a parameter that controls the relative strength of the prior (Hubbell, 1965). In the limit $\alpha \to 1$, equation (5) reduces to the principal eigenvector problem

$$\boldsymbol{\pi}' \;=\; \mathbf{H}^\top \boldsymbol{\pi}', \qquad \mathbf{1}^\top \boldsymbol{\pi}' = 1, \quad \boldsymbol{\pi}' \geq 0, \tag{6}$$

whose solution is unique when $\mathbf{H}$ is primitive (irreducible and aperiodic). Geller (1978) shows that $\boldsymbol{\pi}'$ is then the stationary distribution of the Markov chain with transition matrix $\mathbf{H}$, and that it coincides with the *influence weight* of Pinski and Narin (1976) scaled by the reference count proportion $r_p / \sum_q r_q$.

The spectral ranking $\boldsymbol{\pi}$ reflects the total influence of each unit and depends on unit size. The *influence per work* (Palacios-Huerta and Volij, 2004),

$$\mathbf{v} \;=\; A\, \mathbf{A}^{-1} \boldsymbol{\pi}, \qquad \mathbf{A} = \operatorname{diag}(a_p), \qquad A = \mathbf{a}^\top \mathbf{1} = \textstyle\sum_p a_p, \tag{7}$$

removes this dependence, where the unit work count $a_p = |\{i : w_i \in p\}|$ where p is a specific source or institution, $\boldsymbol{\pi}$ denotes the solution to (5), and $\boldsymbol{\pi}'$ the limiting case (6). The scalar A normalises $\mathbf{v}$ so that its a_p-weighted mean equals one: $\mathbf{a}^\top \mathbf{v} / A = \mathbf{1}^\top \boldsymbol{\pi} = 1$. A unit with $v_p = 1$ therefore has the corpus-average influence per work and $v_p = 2$ indicates twice the average influence per work. Palacios-Huerta and Volij (2004) show that $\mathbf{v}$ is uniquely characterised by several reasonable axiomatic constraints (but see Serrano, 2004).

To compute $\boldsymbol{\pi}$ we define the modified matrix

$$\tilde{\mathbf{H}} \;=\; \alpha \mathbf{H} + (1-\alpha)\boldsymbol{\mu}\mathbf{1}^\top, \tag{8}$$

so that the fixed-point condition $\boldsymbol{\pi} = \tilde{\mathbf{H}}^\top \boldsymbol{\pi}$ is equivalent to equation (5). Because $\boldsymbol{\mu} > 0$ and $0 < \alpha < 1$, every entry of $\tilde{\mathbf{H}}$ is strictly positive, ensuring it is a primitive row-stochastic matrix. By the Perron–Frobenius theorem, $\tilde{\mathbf{H}}^\top$ therefore has a unique stationary distribution $\boldsymbol{\pi}$, recovered by power iteration

$$\boldsymbol{\pi}^{(k+1)} \;=\; \tilde{\mathbf{H}}^\top \boldsymbol{\pi}^{(k)}, \tag{9}$$

converging to $\boldsymbol{\pi}$ from any strictly positive starting vector $\boldsymbol{\pi}^{(0)}$. Note that the conversion from total to per-work influence by the scaling $A\,\mathbf{A}^{-1}$ must be applied after π has converged.

Bipartite ranking. With $\mathbf{m} = (0110)$, equation (4) yields block off-diagonal matrices z$\mathbf{C}$ and $\mathbf{H}$, forming a directed bipartite network between sources (S) and institutions (I). Partitioning rows and columns into the S and I blocks, the stochastic matrix takes the form

$$\mathbf{H} \;=\; \begin{pmatrix} \mathbf{0} & \mathbf{H}_{SI} \\ \mathbf{H}_{IS} & \mathbf{0} \end{pmatrix}, \tag{10}$$

where $\mathbf{H}_{SI}$ distributes a source's attention across institutions and $\mathbf{H}_{IS}$ does the reverse. Because every path alternates $S \to I \to S \to \cdots$, this structure makes $\mathbf{H}$ periodic with period 2 and the undamped eigenvector equation (6) has no unique solution. The modification $\tilde{\mathbf{H}}$ in equation (8) restores primitivity for any $\alpha \in (0, 1)$.

Squaring $\mathbf{H}$ collapses the two-step walks back onto each node type, giving the block-diagonal one-mode projections

$$\mathbf{H}^2 \;=\; \begin{pmatrix} \mathbf{M}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_I \end{pmatrix}, \qquad \mathbf{M}_S = \mathbf{H}_{SI}\mathbf{H}_{IS}, \quad \mathbf{M}_I = \mathbf{H}_{IS}\mathbf{H}_{SI}.$$

Both $\mathbf{M}_S$ and $\mathbf{M}_I$ are row-stochastic products of row-stochastic matrices.

To make the bipartite mode directly comparable with the unipartite SS and II modes at the same α, each single arc SI or IS is damped by $\sqrt{\alpha}$, so every round-trip through $\mathbf{M}_S$ incurs attenuation $\sqrt{\alpha} \cdot \sqrt{\alpha} = \alpha$, matching the single-step damping elsewhere. Substituting $\sqrt{\alpha}$ for α in equation (6), partitioning $\boldsymbol{\pi} = (\boldsymbol{\pi}_S, \boldsymbol{\pi}_I)$, and eliminating $\boldsymbol{\pi}_I$ yields the bipartite resolvent for sources,

$$\big(\mathbf{I} - \alpha \mathbf{M}_S^\top\big)\boldsymbol{\pi}_S \;=\; (1 - \sqrt{\alpha})\big(\boldsymbol{\mu}_S + \sqrt{\alpha}\,\mathbf{H}_{IS}^\top \boldsymbol{\mu}_I\big), \tag{11}$$

where the right-hand side transports institution prior mass back to sources through $\mathbf{H}_{IS}^\top$. The institution scores are then recovered by a single matrix–vector product,

$$\boldsymbol{\pi}_I \;=\; \sqrt{\alpha}\,\mathbf{H}_{SI}^\top \boldsymbol{\pi}_S + (1 - \sqrt{\alpha})\boldsymbol{\mu}_I. \tag{12}$$

$\mathbf{C}(\chi, \mathbf{m}, \rho)$ is a sparse matrix built from a weighted edge list $\mathcal{E}$ accumulated over $\mathfrak{R}$ with per-reference weight ρ_i, stored in CSR format, and row-normalised to $\mathbf{H}$. The pipeline $\mathfrak{R} \to \mathcal{E} \to \mathbf{C} \to \mathbf{H} \to \boldsymbol{\pi} \to \mathbf{v}$ is implemented using `scipy.sparse.csr_matrix` with parameter entry points detailed in the Supplement.

3 Corpus selection

OpenAlex (Priem et al, 2022) provides our bibliometric data. Corpus selection starts with a list of eligible sources having four features: an OpenAlex source identifier for corpus extraction, evidence from one of our journal registries that the source publishes economics or business research, a sufficient number of published works per year, and relatively high OpenAlex source topic density to exclude cross-disciplinary journals.

The journal registries are as follows. Harzing's *71st Edition of the Journal Quality List* (Harzing, 2024, JQL) assembles peer assessments of journal quality across business and economics disciplines. Its scope is confined to economics and business so no filtering is applied. The *ERA 2023 Submission Journal List* (Australian Research Council, 2022, SJL) assigns up to three Field of Research (FoR) codes to each entry. From the full list of 26,241 journals we retain those having all FoR codes falling within Division 35 (Commerce, Management, Tourism and Services) or Division 38 (Economics). This applies to two-digit 'division' codes and all four-digit 'group' codes (3501–3509, 3801–3803, 3899). After filtering, 1,602 journals remain. *Clarivate Web of Science Journal Citation Reports* (Clarivate Analytics, 2026, JCR) assign each journal a Web of Science category. We retain those in Economics, Management, Business, Transportation or Business, Finance. After filtering, 1,413 journals remain.

Every journal in JQL, SJL, and MJL has at least one ISSN, and may have several. These registry ISSNs are joined independently to the February 2026 OpenAlex sources snapshot using its list of ISSNs as the matching key and selecting only the 'journal' source-type. Journal titles are examined to identify around 30 additional matches where the OpenAlex source does not have an ISSN. The list of OpenAlex sources that are matched is designated OAS*.

Table 1: Matching of the three register-based sources to OpenAlex source identifiers. Overlaps count shared matched entries; OAS* is the union across all three registries.

Register	Journals	Matched	Match rate	Unmatched
JQL	842	837	99.4%	5
MJL	1,413	1,392	98.5%	21
SJL	1,602	1,369	85.5%	233
Overlap: MJL $\cap$ SJL				782 sources
Overlap: MJL $\cap$ JQL				567 sources
Overlap: SJL $\cap$ JQL				422 sources
Overlap: MJL $\cap$ SJL $\cap$ JQL				393 sources
(JQL $\cup$ MJL $\cup$ SJL) $\cap$ OA (OAS*)				2,359 sources

The long list OAS* includes sources publishing many works in disciplines beyond economics and business, and we use OpenAlex source-topic assignments to exclude them as follows. A source has work-topic counts for up to five topics, each within a single sub-field, field, and domain. We accumulate the work-topic counts for OpenAlex Field 14 (Economics, Econometrics and Finance) and Field 20 (Business and Management) and drop sources if this count is below 50% of the total count.

For the surviving OAS* sources we extract work and authorship data revealing units that can be excluded because they publish very few works. To find a suitable threshold for excluding these units, we examine source and institution retention curves (Fig. 1), selecting thresholds of $\tau_U = 20$ works/institution/year and $\tau_S = 20$ works/source/year. We designate the residual set of sources OAS.

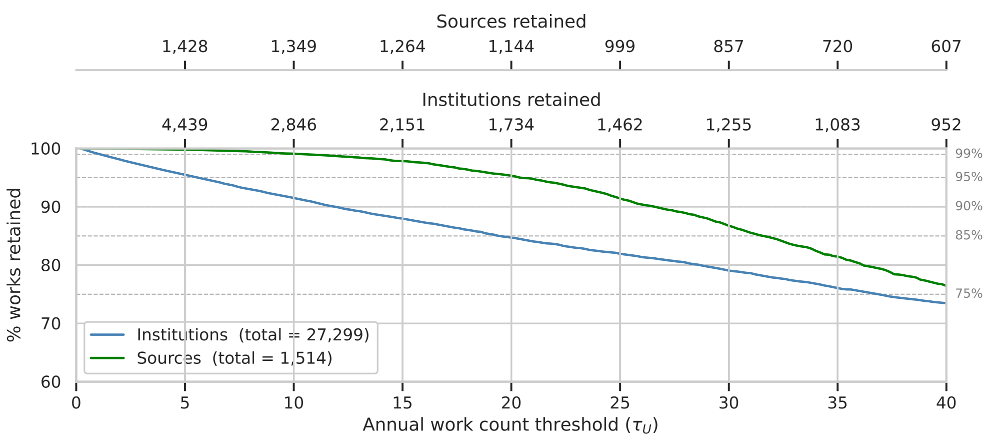

Figure 1: Institution and source retention curves

Using units that satisfy these thresholds, we extract a full corpus of works, authorships and references retaining article and review types with publication years from 2000 to 2024, excluding works that are paratext or retracted, or without an authorship entity, and compute ω_{iu} for each work. The characteristics of the full corpus are exhibited in Table 2.

For the bipartite mode, the one-mode projection $M_S = H_{SI}H_{IS}$ represents a two-hop walk $S \to I \to S$, so the damping applied to M_S in power iteration is α^2, ensuring that the per-hop attenuation is α in all

Table 2: Corpus features by year ($\tau_U > 20$). Upper panel: unique-item counts. Lower panel: D_1 and D_9 deciles of per-entity distributions. The 2000–24 column pools the full period.

	2000		2024		2000–24	
Quantity	count		count		count	
Works	15,011		51,902		860,101	
Sources	664		1,307		1,610	
Institutions	875		1,010		1,012	
Reference counts (out-degree)	5,997		1,378,383		13,590,934	
Citation counts (in-degree)	559,649		28,613		14,036,195	
Quantity	D_1	D_9	D_1	D_9	D_1	D_9
Works per source	3.0	46.0	4.0	78.0	26.9	1191.2
Institutions per work	1.0	3.0	1.0	5.0	1.0	4.0
References per work (out-degree)	1.0	3.0	5.0	60.0	2.0	44.0
Citations per work (in-degree)	2.0	95.0	1.0	4.0	1.0	47.0

modes and that v_s scores are directly comparable across the source-only and bipartite rankings at the same α.

4 Results

4.1 Network modes and analysis sequence

Each corpus extraction is parameterised by a field subset F and a time window t_x. The field subset restricts the source set $\mathfrak{S}$ to sources whose OpenAlex topic counts fall exclusively in Field 14 ($F = E$), exclusively in Field 20 ($F = B$), or in either field ($F = A$, all OAS sources). The time window encodes the census and target year ranges via $t_x \in \{1, \dots, 7\}$; the seven cases are listed in Table 4. The five symmetric windows correspond to conventional five-year citation windows; cases 6 and 7 are asymmetric windows analogous to the Journal Impact Factor and reference spectroscopy respectively.

The corpus comprises N_s sources and N_u institutions linked by a reference network whose attention flow decomposes into four block types (SS, SI, IS, II) as defined in Section 2. The block mask $\mathbf{m} \in \{0, 1\}^4$ selects which blocks enter the assembled citation matrix; the mixing weight χ governs the source–institution balance in cases where both unit types are active. Four choices of $\mathbf{m}$ are bibliometrically meaningful, and the results are organised around them in sequence.

SS mode $\mathbf{m} = (1, 0, 0, 0)$: source ranking in isolation. With only the source block active, the ranking reduces to a Katz–Hubbell prestige index on the journal-to-journal reference network, the setting most familiar from prior bibliometric work (e.g. Pinski and Narin, 1976; Palacios-Huerta and Volij, 2004). In this mode, Economics and Business journals form well-separated citation communities. The second eigenvector $\boldsymbol{\phi}_2$ of the ranking operator $\mathbf{H}_{SS}^\top$ encodes this partition: journals with strongly positive ϕ_2 cluster in one field, strongly negative in the other, and finance journals appear near $\phi_2 = 0$ as disciplinary bridges drawing prestige from both sides. The spectral gap $g = 1 - \lambda_2$ and the community amplification factor $1/(1 - \alpha\lambda_2)$ quantify how strongly field membership shapes the ranking. SS results provide the reference point against which the institution modes are assessed.

II mode $\mathbf{m} = (0, 0, 0, 1)$: institution ranking in isolation. With only the institution block active, the ranking measures the prestige of research environments rather than publication venues. The community structure visible in the SS mode largely dissolves: multi-faculty universities simultaneously host Economics and Business researchers, so institution-to-institution references cross disciplinary lines far more freely than journal-to-journal references do. The spectral gap of $\mathbf{H}_{II}^\top$ is accordingly larger than that of $\mathbf{H}_{SS}^\top$, and the

Table 3: Model parameters $\boldsymbol{\theta}$.

Symbol	Name	Type	Role
Corpus construction			
$F \in \{E, B, A\}$	field subset	category	restrict $\mathfrak{S}$ to Economics (E), Business (B), or all OAS (A); see Table 4
$t_x \in \{1, \ldots, 7\}$	time window	integer	census and target year ranges; see Table 4
τ_U	institution threshold	integer	minimum mean annual works to retain an institution
τ_S	source threshold	integer	minimum mean annual works to retain a source
Matrix construction			
$\rho \in \{0, 1\}$	reference normalisation	binary	0: a work contributes $\bar{R}$ units 1: a work contributes R_i units
$\mathbf{m} \in \{0, 1\}^4$	block mask	binary 4-vector	select active unit-type pairs (SS, SI, IS, II)
$\chi \in [0, 1]$	mixing weight	continuous	source–institution balance within each work
Ranking			
$\alpha \in [0, 1)$	damping factor	continuous	controls propagation through citation network ensures convergence

amplification of field effects on rankings is correspondingly weaker. Comparing SS and II therefore separates what is specific to the journal channel from what is attributable to the institutional environment.

Full joint mode $\mathbf{m} = (1, 1, 1, 1)$: blending source and institution flows. All four blocks are assembled into a single $N \times N$ matrix with χ controlling the blend. At $\chi = 0$ the full joint collapses to the SS mode; at $\chi = 1$ to the II mode; intermediate values interpolate. The baseline value $\chi = 0.5$ allocates equal total citation flow to each mode, but because $N_u > N_s$ this implies a lower expected prestige per unit for institutions than for sources. The dimensionally neutral calibration

$$\chi^* \;=\; \frac{N_u}{N_s + N_u} \tag{13}$$

corrects this, equalising expected prestige per unit across both types. With $N_s \approx 1{,}600$ and $N_u \approx 2{,}600$ at the baseline corpus, $\chi^* \approx 0.619$. In the full joint case the second eigenvector of the ranking operator may encode the Economics/Business field partition, the source/institution type partition, or a mixture; we examine which dominates at $\chi = 0.5$ and $\chi = \chi^*$.

SI/IS mode $\mathbf{m} = (0, 1, 1, 0)$: the bipartite baseline. Removing the direct SS and II blocks retains only cross-type flows: sources accumulate prestige exclusively through the institutions with which their works are affiliated, and institutions exclusively through the sources associated with them. This is not an intermediate point on the SS-to-II spectrum but a structurally distinct architecture. The one-mode projection $\mathbf{M}_S = \mathbf{H}_{SI}\mathbf{H}_{IS}$ eliminates institutions analytically (Section 2), and its second eigenvector encodes the residual community structure after institution-mediated mixing — attenuated relative to the SS case because each source now communicates with others only via its specific institutional affiliates, which act as partial community mixers. The SI/IS mode is the paper's primary baseline.

Robustness parameters. The reference-count normalisation ρ, the damping factor α, the institution work-count threshold τ_U, and the time window t_x govern details of corpus construction and prestige propagation rather than the architecture of the citation network. Baseline values and the full set of values explored are listed in Table 5; one-at-a-time sensitivity analysis is reported in Part E. Readers primarily interested in the empirical rankings may read Parts A–D independently.

Table 4: Corpus construction parameters: field subsets (F) and time windows (t_x).

Panel A : Field subsets (F)

F	Sources included in $\mathfrak{S}$
E	Topic counts exclusively in Field 14 (Economics, Econometrics and Finance)
B	Topic counts exclusively in Field 20 (Business, Management and Accounting)
A	All OAS sources; topic counts in Fields 14 and/or 20 *(baseline)*

Panel B : Time windows (t_x)

t_x	Census window $[t_0^c, t_1^c]$	Target window $[t_0^t, t_1^t]$	Label
1	2000–2004	2000–2004	Symmetric 5-year
2	2005–2009	2005–2009	Symmetric 5-year
3	2010–2014	2010–2014	Symmetric 5-year
4	2015–2019	2015–2019	Symmetric 5-year
5	2020–2024	2020–2024	Symmetric 5-year *(baseline)*
6	2024	2000–2024	JIF analogue
7	2020–2024	2020	Reference spectroscopy

Table 5: Parameters and values explored. Baseline values in italics.

Symbol	Name	Baseline	Values explored
Corpus construction (see Tables 3 and 4)			
t_x	Time window	*5*	1, 2, 3, 4, *5*, 6
F	Field subset	A	E, B, A
τ_U	Institution threshold	*10*	8, *10*
Matrix construction			
ρ	Reference normalisation	*0 (fixed)*	*0* (fixed), 1 (full)
$\mathbf{m}$	Block mask	$(0,1,1,0)$	$(1,0,0,0)$, $(0,0,0,1)$, $(0,1,1,0)$, $(1,1,1,1)$
χ	Mixing weight	*n/a (bipartite)*	*0.50*, χ^* [full joint only]
Ranking			
α	Damping factor	*0.85*	0.50, *0.85*

4.2 Part A — Source ranking: SS mode

[Placeholder: Tables S1–S3 (top sources SS F=A/E/B), rank-shift exhibit, community partition Table S4, Figure 3 Panel A.]

4.3 Part B — Institution ranking: II mode

[Placeholder: Table S5 (top institutions II F=A), Figure 3 Panel B, SS vs II community comparison.]

4.4 Part C — Bipartite SI/IS baseline

[Placeholder: Tables 1–2 (primary source and institution rankings), Figure 2 updated, rank correlations SS vs SI/IS and II vs SI/IS, Figure 3 Panel C (spectral gap summary).]

4.5 Part D — Full joint mode and χ sensitivity

[Placeholder: Tables S6–S7 (full joint $\chi = 0.5$), Figure 3 Panel D (ϕ_2 at $\chi = 0.5$ vs χ^*), rank correlations.]

4.6 Part E — Parameter sensitivity

[Placeholder: Table S8 (Spearman ρ summary, all Stage 1 variants vs bipartite baseline). Note on $\rho = 0$ vs $\rho = 1$.]

4.7 Part F — Time series stability

[Placeholder: Figure 4 (bump chart top-20 sources across $t_x = 1$–6), Table S9 (pairwise Spearman ρ matrix).]

5 Discussion

Face validity. [Placeholder.]

Content validity. [Placeholder.]

Criterion validity — concurrent and predictive. [Placeholder.]

Construct validity. [Placeholder.]

6 Prospects

[Placeholder.]

7 Addenda

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